# COUNTABILITY OF COHOMOLOGY AND EXT AGAINST A PERFECT OBJECT FOR DERIVED QUASI-COHERENT OBJECTS WITH COUNTABLE SECTION SETS

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file perfect.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

*Allowed standard prerequisites (editorial).*

- Ext groups against a perfect complex are computed as cohomology of the tensor product with the dual perfect complex. invoked at proof lines 8522-8524. Reduces the Ext assertion to the cohomology assertion.
- On an affine scheme the derived category of quasi-coherent modules compares with the derived category of modules, perfect objects correspond to bounded complexes of finite projective modules, and the tensor product of derived quasi-coherent objects is represented by the total complex of the tensor product double complex. invoked at proof lines 8529-8539. Translates the affine case into countability of cohomology of module complexes.
- The induction principle for quasi-compact opens and the unbounded Mayer-Vietoris sequence for cohomology. invoked at proof lines 8525-8526 and proof lines 8549-8553. Provides the gluing step from V, W, and V cap W to U = V cup W.

*Source reference.* Original lemma 08DQ: sources/stacks-chow/tags.txt, source lines 10645–10645. Cited at proof lines 8522-8524 to reduce Ext to cohomology; the affine and gluing arguments are inside the counted span.

*Source reference.* Original lemma 08DR: sources/stacks-chow/tags.txt, source lines 10646–10646. Cited at proof lines 8525-8526 to organize the induction; the affine base case and gluing step are inside the counted span.

*Source reference.* Original lemma 06Z0: sources/stacks-perfect/pinned-original-source.tex, source lines 454–454. Cited at proof lines 8532-8533 to represent K by a module complex; the affine countability argument is inside the counted span.

*Source reference.* Original lemma 08EB: sources/stacks-perfect/pinned-original-source.tex, source lines 2636–2636. Cited at proof lines 8532-8533 to represent E by P^bullet; the induction on its length is inside the counted span.

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*Date*: stacks-project:perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Source reference.* Original definition 0657: sources/stacks-chow/tags.txt, source lines 7891–7891. Cited at proof lines 8534-8535 for the module-level definition used in the representation; the affine argument is inside the counted span.

*Source reference.* Original lemma 08DX: sources/stacks-perfect/pinned-original-source.tex, source lines 620–620. Cited at proof lines 8536-8539 to represent the tensor product by the total complex; the countability argument on it is inside the counted span.

*Source reference.* Original lemma 08BX: sources/stacks-chow/tags.txt, source lines 10582–10582. Cited at proof lines 8552-8553 for the gluing step; the conclusion of the lemma is inside the counted span.

**Lemma 0.2.** *Let $X$ be a quasi-compact and quasi-separated scheme. Let $K$ be an object of $D_{QCoh}(\mathcal{O}_X)$ such that the cohomology sheaves $H^i(K)$ have countable sets of sections over affine opens. Then for any quasi-compact open $U \subset X$ and any perfect object $E$ in $D(\mathcal{O}_X)$ the sets*

$$H^i(U, K \otimes^{\mathbf{L}} E), \quad \mathrm{Ext}^i(E|_U, K|_U)$$

*are countable.*

*Proof.* Using Cohomology, Lemma 08DQ we see that it suffices to prove the result for the groups $H^i(U, K \otimes^{\mathbf{L}} E)$. We will use the induction principle to prove the lemma, see Cohomology of Schemes, Lemma 08DR.

First we show that it holds when $U = \mathrm{Spec}(A)$ is affine. Namely, we can represent $K$ by a complex of $A$-modules $K^\bullet$ and $E$ by a finite complex of finite projective $A$-modules $P^\bullet$. See Lemmas 06Z0 and 08EB and our definition of perfect complexes of $A$-modules (More on Algebra, Definition 0657). Then $(E \otimes^{\mathbf{L}} K)|_U$ is represented by the total complex associated to the double complex $P^\bullet \otimes_A K^\bullet$ (Lemma 08DX). Using induction on the length of the complex $P^\bullet$ (or using a suitable spectral sequence) we see that it suffices to show that $H^i(P^a \otimes_A K^\bullet)$ is countable for each $a$. Since $P^a$ is a direct summand of $A^{\oplus n}$ for some $n$ this follows from the assumption that the cohomology group $H^i(K^\bullet)$ is countable.

To finish the proof it suffices to show: if $U = V \cup W$ and the result holds for $V$, $W$, and $V \cap W$, then the result holds for $U$. This is an immediate consequence of the Mayer-Vietoris sequence, see Cohomology, Lemma 08BX. $\square$